

DESIGN AND DEPLOYMENT OF A SECURE CLOUD-BASED PATIENT DATA
MANAGEMENT SYSTEM USING A THREE-TIER ARCHITECTURE ON AWS

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DESIGN AND DEPLOYMENT OF A SECURE CLOUD-BASED PATIENT DATA
MANAGEMENT SYSTEM USING A THREE-TIER ARCHITECTURE ON AWS

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A thesis submitted in fulfilment of the
requirements for the award of the degree of
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I dedicate this template to those who have spent countless hours to make this possible.

ABSTRACT

In this study, a design of the Secure Cloud-Based Patient Data Management System for the Alamin Clinic of Saudi Arabia, a privately owned healthcare organization, is introduced. This private healthcare facility faced the challenge of having its on-premise infrastructure breached by a ransomware that led to the encryption of all patient data and operational disruptions. Four major vulnerabilities that have allowed such cyber-attack are identified in this study. They include a flat network architecture; manual infrastructure management ; lack of data encryption, and a lack of infrastructure recovery plan. Thus, these factors enabled the ransomware attack. They also reflect the security posture of other similar small scale private healthcare organizations that use their legacy on-premises infrastructure. In order to address these challenges, the proposed design uses a three-tier architecture hosted within the environment of Amazon Web Services, where presentation, application, and database layers are placed in separate subnets within a VPC and database tier has no Internet access paths. In addition, a solution with Infrastructure as Code is used using Terraform. It allows not only an automated process of deployment but also quick recovery in case of infrastructure crashes. The DevSecOps continuous integration and delivery pipeline uses Trivy, SonarQube, and Checkov applications and provides automated vulnerability scan at each code commit, making any critical finding impossible to deploy. Finally, the access to the application and database is restricted using Identity and Access Management policies and Role Based Access Control. Row-level security of the PostgreSQL databases ensures the enforcement of individual data isolation at the storage layer. The proposed network topology, security group configuration, database schema with row-level security policies, DevSecOps pipeline specifications, and wireframe for interfaces are included in this design to establish an architectural blueprint. The main evaluation criteria are set as a recovery time objective under fifteen minutes and achieving passing HIPAA security posture, measured through AWS Security Hub.

ABSTRAK

Dalam kertas penyelidikan ini, reka bentuk Sistem Pengurusan Data Pesakit Berasaskan Awan yang Selamat untuk Klinik Alamin di Arab Saudi, sebuah organisasi penjagaan kesihatan swasta, diperkenalkan. Fasiliti ini menghadapi cabaran pencerobohan infrastruktur on-premise oleh perisian tebusan yang mengakibatkan penyulitan data pesakit dan gangguan operasi. Empat kerentanan utama dikenal pasti: seni bina rangkaian rata, pengurusan infrastruktur manual, ketiadaan penyulitan data, dan ketiadaan pelan pemulihan. Faktor ini membolehkan serangan tersebut dan mencerminkan tahap keselamatan organisasi kesihatan kecil lain yang menggunakan infrastruktur warisan. Bagi menangani cabaran ini, reka bentuk cadangan menggunakan seni bina tiga lapis pada Amazon Web Services, di mana lapisan persembahan, aplikasi, dan pangkalan data diasingkan dalam subnet VPC tanpa capaian Internet bagi pangkalan data. Infrastruktur sebagai Kod menggunakan Terraform membolehkan pelaksanaan automatik dan pemulihan pantas. Talian paip DevSecOps menyepadukan Trivy, SonarQube, dan Checkov untuk imbasan kerentanan automatik pada setiap commit bagi menghalang penggunaan kod berisiko. Capaian dihadkan melalui polisi IAM dan RBAC, manakala keselamatan peringkat baris PostgreSQL memastikan pengasingan data pada lapisan storan. Topologi rangkaian, konfigurasi kumpulan keselamatan, skema pangkalan data, spesifikasi talian paip DevSecOps, dan rangka dawai antara muka disertakan sebagai pelan tindakan pelaksanaan PSM2. Kriteria penilaian utama adalah objektif masa pemulihan bawah lima belas minit dan pencapaian tahap keselamatan HIPAA melalui AWS Security Hub.

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LIST OF ABBREVIATIONS

AES	-	Advanced Encryption Standard
ALB	-	Application Load Balancer
API	-	Application Programming Interface
AWS	-	Amazon Web Services
AZ	-	Availability Zone
CI/CD	-	Continuous Integration and Continuous Delivery
CIDR	-	Classless Inter-Domain Routing
CVE	-	Common Vulnerabilities and Exposures
DevSecOps	-	Development, Security, and Operations
EBS	-	Elastic Block Store
ECR	-	Elastic Container Registry
EHR	-	Electronic Health Record
ER	-	Entity-Relationship
FHIR	-	Fast Healthcare Interoperability Resources
FR	-	Functional Requirement
FTP	-	File Transfer Protocol
HCL	-	HashiCorp Configuration Language
HIPAA	-	Health Insurance Portability and Accountability Act
HL7	-	Health Level 7
HMS	-	Hospital Management System
HTTPS	-	Hypertext Transfer Protocol Secure
IAM	-	Identity and Access Management
IaC	-	Infrastructure as Code
JWT	-	JSON Web Token
KMS	-	Key Management Service
NACL	-	Network Access Control List
NAT	-	Network Address Translation

NFR	-	Non-Functional Requirement
NVD	-	National Vulnerability Database
PC	-	Personal Computer
PDMS	-	Patient Data Management System
PDPL	-	Personal Data Protection Law
PSM	-	Projek Sarjana Muda
RBAC	-	Role-Based Access Control
RDS	-	Relational Database Service
RLS	-	Row-Level Security
RTO	-	Recovery Time Objective
SAST	-	Static Application Security Testing
SHA	-	Secure Hash Algorithm
SQL	-	Structured Query Language
SSL	-	Secure Sockets Layer
TLS	-	Transport Layer Security
UAT	-	User Acceptance Testing
UC	-	Use Case
UML	-	Unified Modelling Language
US	-	User Story
UTM	-	Universiti Teknologi Malaysia
UUID	-	Universally Unique Identifier
VPC	-	Virtual Private Cloud
YAML	-	YAML Ain't Markup Language

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The rapid adoption of digital technologies has transformed the entire process of collecting and storing information in the clinic. Transitioning from paper-based processes to the use of EHRs in clinics and hospitals which made healthcare information management infrastructure crucial since sensitive patient data must be protected. Information security and protection of secret patient information has always been an area of concern, especially in cases of breaches where patient privacy and security is put at risk.

While the importance of adopting a secure infrastructure becomes obvious with the growth in the number of cybersecurity threats, there are still many small clinics and hospitals which rely on the legacy on-premises server solution and ignore the growing risks linked with the lack of enough cybersecurity measures. The traditional server infrastructure does not allow implementing any additional measures to prevent attacks or limit their impact on the system, thus increasing the risks Significantly.

Thus, this project aims at designing and implementing a cloud-based solution that would be able to effectively protect patients' confidential information from malicious attacks and unauthorized access. The proposed cloud-based system will feature a three-tier architecture and be deployed into isolated network subnets using Virtual Private Cloud (VPC) and Infrastructure as Code (IaC) solutions such as Terraform. The DevSecOps pipeline will enable developers to scan every piece of code before deployment.

A practical example of the need for this system is seen at the case of the Alamin Clinic which suffered a cyberattack and became a victim of ransomware.

Thus, the selected organization will provide requirements for the system design and implementation to solve the identified problem.

1.2 Problem Statement

At present, the management of the patient data management system at the Alamin Clinic relies on an on-premises physical server that provides services of the web frontend, application backend, and patient database in a single network without any segmentation of layers of the architecture. In other words, all of these elements of infrastructure are manually configured, deployed, and managed; for example, the code for applications is deployed to a production environment through FTP or USB drive while updates for antivirus programs and firewalls are applied on a manual basis with significant delays.

However, such an approach revealed its flaws when the clinic was hacked and its patients' data was affected by the ransomware threat. The hackers gained access to the system, Found its way into the network and encrypted the entire database. Due to the lack of network segmentation and isolation, all the components of the system became vulnerable and unusable until the data is restored manually. The lack of automation and regular backups increased downtime and made it impossible to restore data at time.

It appears that the problem described above align with a wider problem that affects small healthcare organizations. Argaw *et al.* (2019) noted that cyberattacks in hospitals take advantage of old infrastructure, poor network segregation, and inadequate patching policies. Moreover, as Al-Issa *et al.* (2019) reported, cloud computing technologies used in the healthcare industry have some specific problems related to access management and Unprotected date, among others.

In summary, the problems described above reveal one major gap typical for small scale organizations that is a lack of proper information security measures implemented since the beginning. To address the issue, the proposed solution will be based on cloud computing and ensure security from the very beginning through Compliance with the security-by-design principle.

1.3 Project Aim

The aim of this project is to design and deploy a secure cloud-based Patient Data Management System for Alamin Clinic using a three-tier architecture on AWS, Integrating Infrastructure as Code and a DevSecOps pipeline to ensure automated, reproducible, and security-validated deployments.

1.4 Project Objectives

The objectives of the project are:

(a) To investigate core concepts including cloud computing security, three-tier architecture, network segmentation, Identity and Access Management, and secure system design practices within the context of healthcare applications.

(b) To develop a secure Cloud-Based Patient Data Management System based on a three-tier architecture on AWS, covering Virtual Private Cloud networking, access control through IAM and Role-Based Access Control (RBAC), and a DevSecOps CI/CD pipeline with integrated security scanning.

(c) To evaluate the system's performance through automated vulnerability scanning using Trivy, SonarQube, and Checkov. Evaluation includes RTO stress testing via a simulated ransomware recovery scenario and a HIPAA compliance assessment using AWS Security Hub. Finally, User Acceptance Testing (UAT) will be conducted with Doctors, Admins, and Patients to validate functional requirements.

1.5 Project Scope

This project focuses on the design and deployment of a Secure Patient Data Management System for use by doctors, administrative staff, and patients at Alamin Clinic. The system will be deployed on AWS as a functional prototype, with security monitoring through scan reports, CloudWatch logs, and recovery time test results.

1.5.1 In-Scope

The following areas are within the scope of this project:

(a) Core patient record management functionality, including Patient Registration, Medical Records Management, Appointment Scheduling, and User Authentication with Role Management (Doctor, Admin, Patient).

(b) Design of a Virtual Private Cloud (VPC) with public subnets for public internet and private subnets for the application and database layers.

(c) Three-tier architecture Consists of a React frontend layer, a Node.js/Express backend layer on EC2, and a PostgreSQL database layer on Amazon RDS.

(d) AWS services including VPC, EC2, RDS, Application Load Balancer (ALB), NAT Gateway, and Internet Gateway.

(e) Network security controls including Security Groups and Network Access Control Lists (NACLs).

(f) Identity and Access Management (IAM) with least-privilege policies and Role-Based Access Control (RBAC) applied across all three user roles.

(g) Data encryption at rest using AES-256 through AWS Key Management Service (KMS) and encryption in transit using TLS/HTTPS.

(h) A DevSecOps CI/CD pipeline built on GitHub Actions with Docker containerization and Terraform IaC, incorporating automated security scanning using Trivy (container images), SonarQube (static code analysis), and Checkov (IaC misconfiguration scanning).

(i) Monitoring and audit logging using Amazon CloudWatch and AWS CloudTrail.

(j) HIPAA compliance posture assessment using AWS Security Hub.

1.5.2 Data and Subjects

The system will be evaluated using a pilot dataset of simulated patient records generated for testing purposes. User Acceptance Testing will be conducted with a minimum of three representative participants, one per user role (Doctor, Admin, Patient). No real patient data will be used during development or testing.

1.5.3 Out-of-Scope

To maintain focus on the security architecture and infrastructure, the following modules are explicitly excluded from this project:

(i) hospital billing and insurance claim management; (ii) pharmacy inventory and dispensing management; (iii) Emergency Room (ER) management; and (iv) specialized Obstetrics and Gynaecology (O&G) clinical modules.

1.6 Project Importance

The importance of the project comes out in several aspects: clinical, technical, and academic.

First of all, when it comes to clinical issues, one should understand that patient information is considered to be one of the most sensitive types of personal data. An information leak or loss of availability caused by a ransomware attack like in Alamin Clinic's case may have serious consequences, both for patients and doctors, as well as reputational and legal ramifications. Therefore, the development of a system that will be characterized by high availability, security, and integrity of patient data cannot be regarded as a technical task.

Secondly, one needs to consider the technical part of the project. Infrastructure as Code and DevSecOps approaches allowed for the conversion of an outdated and manual environment into a self healing infrastructure with the use of code and security validation. The whole infrastructure is coded using the Terraform language, and this gives us the opportunity to redeploy it from a clean state in minutes. Furthermore, a security scan conducted prior to deployment ensures that any threats are discovered and prevented from moving forward.

Finally, On the academic aspect. In this case, the project allows for practical application of the theoretical notions in terms of cloud security. Using HIPAA as the standard of compliance and AWS Security Hub as an evaluation tool.

1.7 Report Organization

This report is organized as follows:

Chapter 1 — Introduction presents the background to the project, the problem statement at Alamin Clinic, the project aim and objectives, scope, and the importance of the study.

Chapter 2 — Literature Review presents the background literature relevant to the project. Analysis of the current system at Alamin Clinic, a comparison of existing healthcare management systems, and a review of the technologies and methods used in the proposed solution, including cloud security frameworks, three-tier architecture, Infrastructure as Code, and DevSecOps practices.

Chapter 3 — System Development Methodology presents the development approach adopted for this project. It justifies the choice of methodology, outlines its phases as applied to this system, describes the technologies used, and presents the system requirements analysis.

Chapter 4 — Requirement Analysis and Design presents the detailed functional and non-functional requirements of the system, followed by the complete system design including the network architecture, database schema, IAM policy structure, and system interface design.

Chapter 5 — Conclusion summarizes the achievement of the project objectives, reflects on the limitations of the current implementation, and proposes directions for future improvement.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Developing a secure cloud-based system that manages patients' information require knowledge from various areas, which involve threats faced by healthcare information systems, architecture principles used in cloud computing, patients' data regulations, and DevSecOps processes that ensure that the system is developed and deployed in a secure manner. This chapter reviews the knowledge on how the proposed system can be designed are reviewed. This is done through first considering the case study organization before moving to general technological and literature.

Section 2.2 introduces the case study organization, Alamin Clinic, through analysing its organizational structure, current manual operating model, and the interview outcomes from key people in clinic. The section also analyzes the difference between the current followed process and the process followed when the new system is introduced in the clinic. Section 2.3 carries out a 5W 1H analysis on the problems faced by the current system. Section 2.4 compares the proposed system to existing healthcare systems, including the unique aspects from each existing healthcare management system. Section 2.5 highlights the major technologies used in designing the new system and their supporting literature. Section 2.6 presents key conclusions from the literature review.

2.2 Case Study: Alamin Clinic (Saudi Arabia)

Alamin Clinic is a private independent health care service in Saudi Arabia. In essence, this is the organization to be considered a stakeholder and used as the main case study for this proposal. The incident that has happened to Alamin Clinic and led to the development of this idea was an attack by ransomware on their IT services.

The patient base of Alamin Clinic includes people who seek various types of consultations, follow-up sessions, and health record administration services. The key stakeholders, which include doctors, administrative staff, and patients themselves, all have different ways of interacting with the database of patients in the clinic, and therefore, will need different access control mechanisms. This diversity of roles makes the design of a Role-Based Access Control (RBAC) model a fundamental requirement of the proposed system.

2.2.1 Alamin Clinic: Organizational Structure

Alamin Clinic has a three-level organizational hierarchy that is usually common among small private clinics in the region.

The clinical level consists of general specialist doctors who consult patients, diagnose illnesses, and keep medical records. Doctors need read and write permissions on the medical records they treat. They should not be able to access administrative records like bills and salaries of employees.

The administrative level is made up of people who register and schedule patients. Administrators should only be able to access registration data and appointment data but should not be able to see anything else like medical records. Separation of responsibilities is crucial for patient confidentiality and should be guaranteed by the IAM structure proposed here.

Patients registered at Alamin Clinic can log into the application and check their medical records and scheduled appointments. Patient access will only allow them to read their records but not any other data like other patients' records or any administrative record.

The clinic also has a basic IT function that had only on-premise servers. There was no security policy or any patch management policy before the proposed migration.

The organizational structure is illustrated in Figure 2.1.

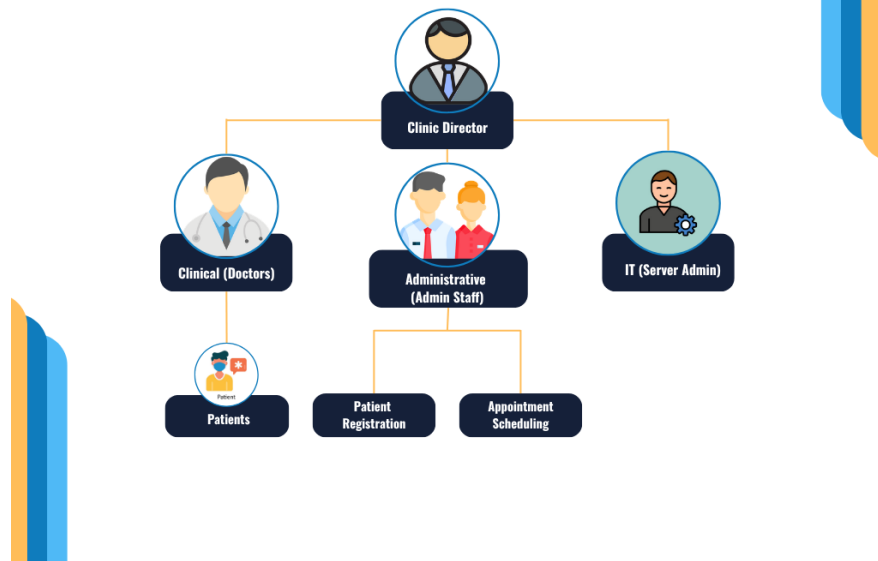


Figure 2.1 Alamin Clinic Organizational Structure

2.2.2 Manual Operation

The existing system used at Alamin Clinic for storing patient information follows a traditional on-premises design model defined by the following characteristics: manual server configuration, manual deployment, local hosting, and reactive security.

Manual Server Configuration. The server hardware installed at the clinic is configured locally by IT staff of the clinic. Server configuration is done manually without maintaining infrastructure versions and automating server deployment processes. This methodology is Unreliable because it relies on manual configuration of servers with no control over drift. It is impossible to roll back the configuration in case of failure because no automated server provisioning is used (Paidy and Chaganti, 2024).

Manual Deployment. The code deployed in the development environment is manually moved to the production server with the use of either FTP clients or storage devices, such as USB drives. This methodology slowdown the deployment time and prevents implementing automatic checks of the code for security or quality issues before deploying it to the production environment. Furthermore, the manual

deployment process ensures that malware can enter the production environment bypassing any controls.

Local Hosting. The frontend of the website, the backend logic of the application, and the patient database are all hosted on the same physical servers. There is no network segmentation between these layers, which means that gaining access to any of the components is enough to access the rest. The absence of network segmentation and isolation is considered one of the key factors that contribute to ransomware infections among healthcare facilities, allowing the malicious code to move directly to the database layer (Argaw *et al.*, 2019; Thamer and Alubady, 2021).

Reactive Security. Firewalls and antivirus solutions are updated manually by IT staff according to available information about existing threats. This practice does not provide enough protection against emerging cyberattacks because updates are carried out reactively and are therefore lagging behind the threat landscape. As a result, known vulnerabilities are patched late, exposing the facility to attacks.

In effect, it was shown by the fact that the clinic's security was compromised after a ransomware attack, where the hackers managed to encrypt all the patients' information, making the records inaccessible for use by the clinic. As such, it was evident that there were no protective measures to block an intruder from reaching their target data.

2.2.3 Interview with Clinic Stakeholders

In order to get first hand information about the problems faced by the clinic and weaknesses of the system, two major stakeholders of the clinic were interviewed in person through a video conferencing. These include one Administrative Staff member who is involved in registering patients and scheduling appointments and a Clinical Staff member who performs routine operations at the clinic. Considering the privacy concerns of the participants, there was no audio recording of the interviews; instead, notes were made while conducting the interviews. There were a total of eight open-ended questions in the interview protocol.

The following questions were used to guide each interview:

- (a) How do you currently access and manage patient records on a daily basis?
- (b) What are the most significant difficulties you face when using the current system?
- (c) How was the ransomware incident discovered, and what was the immediate impact on your work?
- (d) How long was the clinic unable to access patient records, and how was this period managed?
- (e) What data was lost or inaccessible as a result of the attack?
- (f) Were there any security policies or backup procedures in place at the time of the incident?
- (g) What features of a new system would most improve your day-to-day work?
- (h) What security or privacy concerns do you have about moving patient data to a cloud system?

Table 2.1 Summary of Stakeholder Interview Findings and System Implications

Question Theme	Administrative Staff	Clinical Staff	System Design Implication
Daily Access	Individual logins for a 3rd-party desktop app via external vendor. Limited remote access.	Same desktop app; managed by provider. No in-house IT capability for diagnosis.	Transition to a web-based React interface. Remote access supported by default via browser.
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Question Theme	Administrative Staff	Clinical Staff	System Design Implication
Main Difficulties	No unique patient IDs cause duplication. Slow billing. Manual Excel tracking for appointments.	Inconsistent entry compounding duplication. No automated monitoring or alerting for the server.	UUIDs enforced at DB level. Integrated scheduling module replaces Excel. CloudWatch for alerting.
Ransomware Discovery	All data inaccessible (no R/W/M). Operations halted. Used paper forms as fallback.	All departments frozen simultaneously. Doctors could not access files; appointments cancelled.	Private subnet isolation for DB. Security Groups and NACL rules enforce strict tier-based blocking.
Downtime Duration	72 hours of total blackout. Restoration failed due to corrupted or incomplete data.	5-day recovery window. Significant backlog accumulated before operations resumed.	Terraform-based IaC allows full environment rebuild in < 15 mins. Independent RDS snapshots.
Data Lost	All patient records inaccessible. No clean backup existed to restore the system.	Records in the attack period were permanently lost. Access unavailable during recovery.	Daily RDS snapshots with point-in-time recovery. Multi-AZ standby replica for high availability.
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Question Theme	Administrative Staff	Clinical Staff	System Design Implication
Security Measures	Basic antivirus with manual updates. No formal backup policy or scheduled reviews.	No formal security policy or staff awareness of recovery procedures. Systemic procedural gap.	Security Hub continuous HIPAA monitoring. Configurations (IAM, NACLs) version-controlled in Terraform.
Improvements	Web interface, auto-notifications, remote access, unique IDs, and a public website.	Dept-specific modules, automated workflows, and faster performance during consultations.	Role-based dashboards in React. Patient UUIDs for clean registration. Public site is future scope.
Migration Concerns	Data sovereignty and internet reliability in the clinic's operating region.	Data privacy, access control clarity, and training requirements for the new system.	Hosting in AWS Singapore. DB-level Row-Level Security (RLS) ensures users only access authorized records.

The results of the interview confirm the four operational issues found during the interview analysis discussed in Section 2.2.2. However, it brings out three new dimensions to the problem. Firstly, the human frustration on the organization taken by the ransomware attack a five days of operational interruption and full shutdown of clinic operation; secondly, the lack of security policy knowledge and backup practices on the part of operational staff and thirdly, the data quality issue associated with the difficulty to link patients to unique ID numbers.

2.2.4 Current System Workflow

The current process for managing patient information in Alamin Clinic relies entirely on manual handling without any automated means. Figure 2.2 highlights the flowchart for managing the main processes for patient information from the arrival of patients to storing their records.

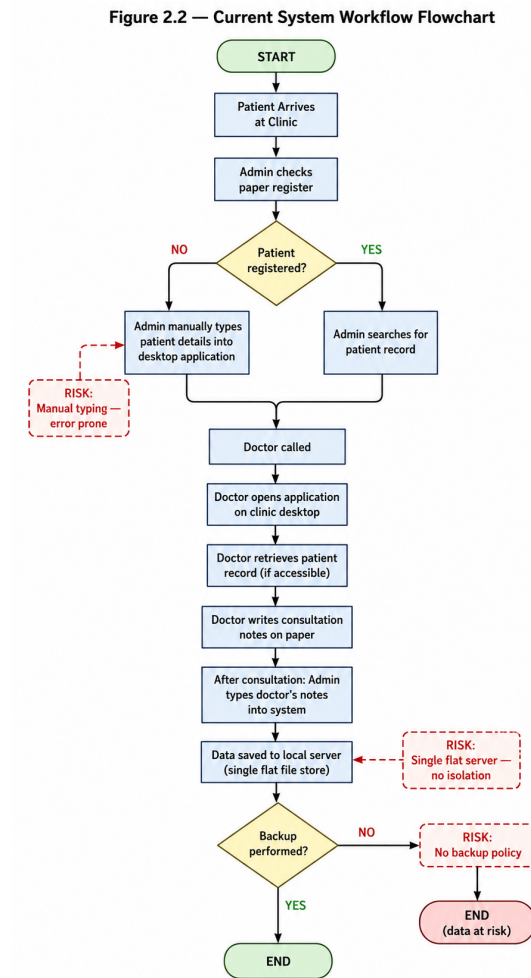


Figure 2.2 Current System Workflow Flowchart

There are five significant threats associated with the current workflow process. Firstly, all inputs into the process are made manually by administrative staff, resulting in errors and time wastage. Secondly, there is no distinction between the application that processes the information and the database that holds the data because both use the same server process and file structure. Thirdly, there is no distinction between the

logins for physicians and administrative staff since both log into the same application with the security measures depending purely on trust. Fourthly, there is no automatic backup system for the process, meaning that all information storage depends entirely on the availability of the actual server. Lastly, there is no way to track who accessed or changed what in patients' files.

2.2.5 Proposed System Workflow

The proposed system replaces each manual, insecure step in the current workflow with an automated, security-enforced equivalent. Figure 2.3 presents the flowchart of the proposed system's patient record management process.

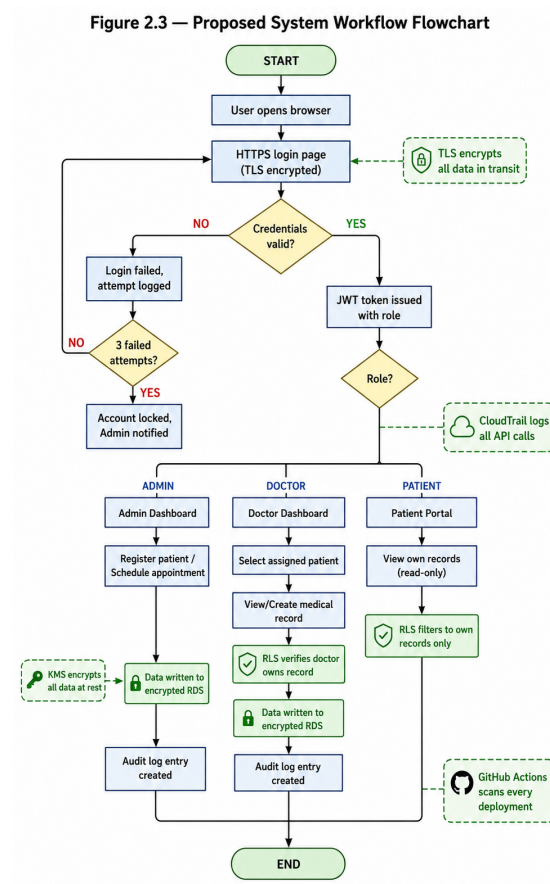


Figure 2.3 Proposed System Workflow Flowchart

The solution presented here solves all of the mentioned vulnerabilities in the system. Instead of manual data entry, there is now an input-validation web page. In

addition, the single-server design is swapped for a three-tier VPC architecture in which the database has been separated into a private subnet that does not have internet access. Instead of using role enforcement, the authentication mechanism relies on JWT and PostgreSQL row-level security to control who accesses what data. Physical hardware backup is no longer used; instead, the system will automatically create snapshots in RDS and redeploy the entire infrastructure with Terraform. Finally, audit logging has been introduced via CloudTrail and an audit table in the application.

Table 2.2 presents a side-by-side comparison of the current and proposed workflows across the five identified vulnerability dimensions.

Table 2.2 Comparison Between the Current System and the Proposed Cloud-Based System

Dimension	Current System	Proposed System
Data entry	Manual typing by admin, error-prone	Structured web forms with validation
Architecture	Single flat server, all tiers co-hosted	Three-tier VPC — presentation, app, DB isolated
Authentication	Single shared login, trust-based	JWT tokens + RBAC enforced at app & DB layers
Backup and recovery	No backup policy; full data loss risk	Automated RDS snapshots + Terraform redeploy in ~ 15 min
Audit trail	None	CloudTrail API logs + application audit_log table
Deployment	Manual FTP/USB, no security gate	GitHub Actions CI/CD with Trivy, SonarQube, & Checkov
Encryption	None at rest; inconsistent in transit	KMS AES-256 at rest; TLS 1.2+ in transit

2.3 Current System Analysis

2.3.1 Problem Analysis Using the 5W 1H Framework

The 5W 1H framework — Who, What, When, Where, Why, and How — provides a structured approach that ensures all dimensions of the issue are addressed before a solution is designed. Table 2.3 applies this framework to the Alamin Clinic problem.

Table 2.3 5W1H Analysis of the Current System Vulnerabilities

Dimension	Analysis
Who is affected?	Doctors cannot access patient records or create diagnoses. Administrative staff cannot manage appointments. Patients cannot receive care during outages. The clinic director faces operational and potential legal liability for data loss.
What is the problem?	A ransomware attack encrypted the patient database. No recovery capability or backups existed. The systemic problem is an infrastructure designed without security controls at any layer.
When does it exist?	At all times. The server is exposed to the internet with no isolation. Patches are reactive, and every manual deployment is a security risk.
Where does it originate?	In the flat network architecture of the on-premise server, providing no barrier to lateral movement. In the absence of network segmentation between web, app, and database layers.
Why does it persist?	Small healthcare providers lack the budget for enterprise security platforms and under-resourced IT staffing. This is the market gap the proposed system addresses.
Continued on next page	

Table 2.3 – Continued from previous page

[HTML]EFEFEF Dimension	Analysis
How will it be solved?	Through a security-by-design cloud architecture: (1) Three-tier VPC isolation. (2) Terraform IaC for configuration consistency. (3) DevSecOps pipeline (Trivy, SonarQube, & Checkov). (4) IAM/RBAC with PostgreSQL security. (5) KMS encryption & TLS in transit. (6) AWS Security Hub for HIPAA compliance.

2.3.2 Security Domain Analysis

The assessment of the existing infrastructure of the Alamin Clinic, before Starting on the migration process, shows that there are weaknesses in the four areas of security:

Network Architecture. In the case of the clinic, the network architecture is highly simplified in that all three tiers of applications—the web, application, and database are hosted in one machine with one network interface, No DMZ for internet connected parts of the network, and no way to prevent lateral movements once the system is breached. This completely violates the concept of defense in depth because it does not offer more than one line of defense (Al-Issa *et al.*, 2019).

Access Control. The user access to the system is determined based on the authentication carried out only at the application level without having any kind of identity management. There are role separations between doctors, administration, and patients that are implemented in the application level but do not have any restrictions at the networking or storage level.

Data Security. The patient data stored in the database of the clinic is not encrypted in its static form. The transmission of the data from the browser of the user to the server is not reliably encrypted using TLS due to the lack of analysis of the HTTP settings used by the clinic. Both the static and the dynamic data are exposed to the threat of being intercepted and stolen(Shojaei *et al.*, 2024).

Operational Resilience. There is no official data backup policy at the clinic, nor is there any officially documented way to recover from such situations. This scenario has been noted in many studies involving ransomware attacks on other small health care providers. Argaw *et al.* (2019) noted that the lack of backup and recovery policy contributed to prolonged disruption during such incidents.

In combination, all of the flaws above shows that the existing model does not meet even the basic criteria of security for a healthcare information system. As such, there is a clear need for a complete overhaul of the system architecture to improve its security level.

2.4 Comparison between existing systems

To frame the proposed model among other solutions for managing healthcare, a comparative table is presented below that shows how the proposed solution fits alongside three other systems, namely, a conventional HMS system, the OpenEMR system, the Epic Systems solution, and the proposed Secure Cloud PDMS.

Table 2.4 Comparison of Existing Healthcare Management Systems

System	Key Features	Security Model	Main Limitations
Traditional On-Premise	Full data control, no internet dependency.	Reactive — manual patches, perimeter firewall.	High ransomware risk, no isolation, poor scalability.
OpenEMR	Patient portal, billing, EHR.	Reactive — community patches.	Manual security updates, no auto-deployment.
Epic Systems	Comprehensive clinical suite, HL7/FHIR.	Proactive — vendor-managed updates, IAM.	Prohibitive cost for small clinics, vendor lock-in.
Proposed System	Patient records, roles, cloud-native.	Proactive, shift-left — automated scanning, RBAC.	Infrastructure focus; billing/pharmacy out of scope.

The comparison highlights a unique weakness of the existing market towards small healthcare providers. On-premise software applications and open-source projects like OpenEMR provide high levels of flexibility but require all aspects of security management to be undertaken by the IT staff of the medical facility, as proven by the history of the Alamin Clinic. However, even though commercial products, including those by Epic Systems, provide an advanced level of security, they are too expensive and complicated for small private providers.

This project will fill this niche through a combination of a low-cost model associated with the use of cloud hosting and the application of automation technologies used in commercial enterprise-level systems. The inclusion of the infrastructure as code through Terraform and the automatic scanning of security issues during the CI/CD process will reduce the need for manual IT procedures, thus minimizing this primary weakness of the existing solution.

2.4.1 Features Adopted from Existing Systems

The comparison provided in Table 2.4 made it possible to choose the features from the existing systems, which would be relevant to the goals of the suggested system both in terms of scope and security considerations. The features borrowed from the existing systems and the modifications made are shown in Table 2.5.

Table 2.5 Features Adopted from Existing Systems

Feature	Adopted From	Justification	Adaptation in Proposed System
Patient Registration	OpenEMR	Core requirement; directly addresses ransomware data loss.	Implemented with PostgreSQL RLS enforcement.
Continued on next page			

Table 2.5 – Continued from previous page

Feature	Adopted From	Justification	Adaptation in Proposed System
Appointment Scheduling	OpenEMR	Operational necessity; appointment data was highly disruptive when lost.	Role-restricted access: Admin modifies, doctors/patients view only.
Patient Portal	OpenEMR	Reduces admin load; addresses patient autonomy.	Read-only access; accounts strictly created by admin.
RBAC Model	Epic Systems	Enterprise-grade pattern validated by Singh & Chatterjee (2015).	Three-layer enforcement: JWT, IAM policies, and PostgreSQL RLS.
Audit Logging	Epic Systems	HIPAA requirement; key missing finding from incident review.	Dual mechanisms: AWS CloudTrail and application audit_log table.
HIPAA Framework	Epic Systems	Recognized benchmark for healthcare data security.	Continuous measurement via AWS Security Hub HIPAA standard.

The features not taken from any current technology are as follows: Billing for hospitals and insurance management (OpenEMR, Epic); Pharmacy management (Epic); ER management (Epic); and HL7/FHIR interoperability (Epic). These features are not incorporated into the scope of the suggested technology to narrow down the scope of the security framework. This is explained in chapter 1, section 1.5 (Project Scope).

Another difference of the system under discussion from its competitors lies in the comparative analysis of the standard AWS architecture design template. Table 2.6 compares these two types of templates.

Table 2.6 Proposed System vs Standard AWS Architecture

Feature	Standard AWS Architecture	Proposed Solution
Deployment	Manual or basic scripts	Fully automated via Terraform IaC
Security Model	Added after build (Reactive)	Shift-left — scanned at every CI/CD stage
Post-attack Recovery	Long manual rebuild process	Instant redeployment from Terraform state
Data Protection	Basic private subnets	Hardened NACLs + Private RDS + KMS encryption
Compliance	Ad hoc manual checks	HIPAA posture measured via AWS Security Hub

2.5 Literature Review of Technology Used

2.5.1 Cloud Computing in Healthcare

Since the early 2010s, the use of cloud computing technology within healthcare has been extensively researched. The study performed by Ahuja *et al.* (2012), which can be considered one of the first comprehensive investigations of cloud computing usage in healthcare settings, concludes that cost savings, scalability, and improved data access are the most common factors contributing to widespread adoption, whereas data protection, regulatory issues, and vendor reliability stand out as the key barriers. All those findings still hold true today. In particular, the security problems mentioned by Ahuja *et al.* (2012), such as concerns about data sovereignty and access control, have influenced the design of the IAM policy for our solution.

In the same spirit, Al-Issa *et al.* (2019) investigated cloud security issues specific to eHealth environments. Their survey revealed that unauthorized access, data corruption, and service availability issues are the leading threats affecting eHealth clouds. Accordingly, each threat mentioned in their study is addressed by the corresponding design requirement in our case: unauthorized access is controlled via IAM/RBAC; integrity violations are prevented using CloudTrail audit logging and KMS encryption; and availability is guaranteed with a multi-AZ architecture and fast recovery provided by Terraform.

The stakeholder concerns regarding cloud data privacy raised by the clinic stakeholders in section 2.2.3 interviews are handled by the proposed solution with the use of (a) the AWS region chosen to be in gulf region according to PDPL data residency rules, (b) IAM least privilege principles, and (c) row-level security of PostgreSQL database which ensures that one cannot access another user's data regardless of infrastructure level access(Ramayanam, 2025).

2.5.2 Three-Tier Architecture

The three-tier architectural approach involves breaking down a web application into three different layers which are physically and logically segregated from one another, namely the presentation tier (frontend), application tier, and data tier (database). There are two key advantages to using this model from a security perspective. First, the damage scope for a breach in a particular tier is significantly reduced. Second, it allows for the independent scaling and access control to be implemented at the level of each individual tier.

In this particular case, the three-tier architecture is deployed using the AWS Virtual Private Cloud (VPC). In accordance with the model, the presentation tier uses an Application Load Balancer running on a public subnet while the application tier uses EC2 instances running on the private application subnet. Finally, the database tier utilizes Amazon RDS, residing in the dedicated private database subnet. The fact that this tier cannot be accessed directly from the Internet can be seen as a built-in protection against the network flatness exploited during the attack at Alamin Clinic.

The proposed design differs from the current design in that it makes use of a three-tiered structure compared to a single-server flat approach used now (Section 2.2.4). The new design includes three tiers that will each have their own separate security group policy, as well as having the database without an outbound Internet route (Gaber *et al.*, 2025).

2.5.3 AWS and the Shared Responsibility Model

The Amazon Web Services runs on a shared responsibility model where there are clearly defined boundaries about who is accountable for what part of the security. It states:

“AWS manages security of the cloud, you are responsible for security in the cloud.”

This means that AWS will take care of the physical security of data centers, the hypervisor level, and the managed services itself. While the customer—the clinic in the case is accountable for OS management, Network Security Groups rules, IAM policies, application security, and data encryption.

It is crucial in the design process for the new system since moving away from traditional hosting to the cloud doesn’t relieve the clinic of any security concerns; instead, it shifts them around. It changes the list of those tasks that have been historically difficult for Alamin Clinic to manage manually to the set that can be easily automated using Terraform, KMS, and IAM policies (Mendoza and Reyes, 2023).

2.5.4 Infrastructure as Code (Terraform)

IaC refers to the use of machine-readable configuration files in lieu of manual procedures to describe and provision computing infrastructure. According to Paidy and Chaganti (2024), IaC significantly decreases configuration drift, making the infrastructure failure recoverability fast and reliable in multi-region AWS deployments—a finding that is directly applicable to the proposed recovery solution.

The chosen IaC tool is HashiCorp's Terraform, which supports a declarative configuration approach that allows the representation of the whole topology of the VPC network, EC2 instance settings, RDS parameter groups, IAM policies, and security group rules in version-controlled code. In effect, the entire environment could be rebuilt from scratch in minutes—an ability that allows turning the Recovery Time Objective into a quantifiable metric.

The discovery directly correlates with the most important insight gained during the stakeholder interview when the IT Administrator was asked about how long it takes to recover from the ransomware attack, and the answer was a period of five days. The recommended RTO target of the system should be less than fifteen minutes, which is possible only due to the configuration being scripted in Terraform and deployable to a clean AWS environment with one command.

Checkov is a static code analyzer tool for IaC that is included in the CI/CD pipeline to scan Terraform scripts before the infrastructure is put to production to avoid common infrastructures security pitfalls, such as excessive permissions on security groups or unprotected storage volumes (Verdet *et al.*, 2023; Espinha Gasiba *et al.*, 2021).

2.5.5 DevSecOps and Shift-Left Security

DevSecOps refers to the process of incorporating security testing and verification during the software development and deployment process, as opposed to mandating security as a gate for the software development and deployment process after development (Valdés-Rodríguez *et al.*, 2023). “Shift-left” involves moving security tests from their current position closer to the development stage, where issues are detected and fixed at the least cost, even before they become part of the production code base (Paidy and Chaganti, 2024).

In the current deployment process, there is no security gate at any stage. This process involves pushing the code to the production server using FTP. In the new system, there is a CI/CD pipeline in GitHub Actions that carries out security scans in three types on each commit:

The proposed CI/CD pipeline integrates three specialized security scanning tools to ensure a "shift-left" security posture:

1. **Trivy:** Scans Docker container images for known Common Vulnerabilities and Exposures (CVEs) before any image is pushed to the container registry.
2. **SonarQube:** Performs Static Application Security Testing (SAST) by analyzing the application source code for security anti-patterns, injection vulnerabilities, and code quality issues.
3. **Checkov:** Analyzes Terraform configuration files for infrastructure-level security misconfigurations, ensuring that the defined AWS resources adhere to best practices

Any pipeline breach during a critical discovery ensures that the software will not be released, guaranteeing that no software containing a critical flaw is ever deployed into the production environment. The move from no security gate to an automated blocking gate in three stages will solve the reactive security stance described in Section 2.2.2(Rajapakse *et al.*, 2022; Akbar *et al.*, 2022).

2.5.6 Identity and Access Management and Role-Based Access Control

IAM refers to the policies, procedures, and technologies that control the access to certain resources by a particular user within the system. IAM on AWS provides granular and rule-based control over access to infrastructure components, thereby complementing the application-based RBAC mechanism.

Singh and Chatterjee (2015) proved that for effective security management in the cloud computing environment, there is a need for security enforcement across various tiers, ranging from the application login page to network and resource levels. This paper's proposed framework follows this approach, as it combines AWS IAM policies, which restrict the API calls available to each application module, with application-level RBAC, which restricts the data that each authorized user can read/write, and PostgreSQL Row Level Security, which enforces isolation of data at the storage level, irrespective of the application level.

Three IAM roles will be created for the intended system according to the clinic's three user groups – Doctor, Admin, and Patient. Specifically, the Doctor role will have read/write access to their patients' information, the Admin role will be allowed to read/write information concerning scheduling and registrations, and the Patient role will only have read access to their own information. Each IAM role will have minimum privileges sufficient for their intended purpose(Butt *et al.*, 2023).

PostgreSQL row-level security as an additional control measure will address the issue raised in Section 2.3 regarding inadequate controls on the access aspect of the current system design since the application-based role access controls offer zero security against direct access to the underlying database, which is what ransomware attackers exploit(Cobrado *et al.*, 2024).

2.5.7 HIPAA Compliance

The Health Insurance Portability and Accountability Act (HIPAA) is the main regulatory framework used in securing and protecting patient health information within the United States and is considered a gold-standard benchmark for healthcare data security across the world. HIPAA Security Rule outlines the technical requirements of access control, audit logging, integrity of data, and encryption during data transmission, all of which relate directly to the IAM, CloudTrail, KMS, and TLS of the proposed system(Abbasi and Smith, 2024).

AWS Security Hub is an AWS-managed service designed to provide continuous posture assessment of the AWS environment in relation to HIPAA compliance and provides detailed assessments where there are discrepancies between the configuration and the standards. The suggested architecture uses AWS Security Hub to measure and assess HIPAA posture compliance.

The current framework does not meet the criteria for any of the technical safeguards prescribed by HIPAA, which includes failure to provide proper access control at the infrastructure level, failure to maintain an audit log of data access activities, absence of data integrity through encryption, and lack of reliable

transmission security measures. The recommended framework caters to all four areas, thus positioning HIPAA readiness as the benchmark for the assessment process.

2.6 Chapter Summary

This chapter provides the theoretical and the comparative groundwork for the design of the proposed Secure Cloud-Based Patient Data Management System.

Through the analysis of the Alamin Clinic case study based on structured interviews with stakeholders, four problems with the current on-premises system have been identified: flat network architecture, poor access controls, lack of encryption of data, and lack of system resilience. With the existing of these four deficiencies together the ransomware attack took place and led to five days of downtime and loss of data. Through the use of the 5W 1H approach, these problems have been categorized along six dimensions.

Workflow analysis suggests that the existing manual, unsecured process may be replaced with the suggested secure automated workflow, which includes: Flat architecture deployed using three layered VPC; FTP deployment is now replaced with CI/CD pipeline for DevSecOps; Roles is set on a basis of trust using JWT and row level security enforcement; redeployment using Terraform in fifteen minutes; and Audit is not conducted due to usage of CloudTrail and logs.

Literature review evidence is provided here to back up both the academic and business community consensus on all chosen core technologies: three-tier VPC design to ensure network isolation, use of Infrastructure as Code through Terraform to ensure that infrastructure is easily repeatable and quickly restorable, use of DevSecOps pipeline to assist in vulnerability detection earlier in the process, multiple access control through IAM/RBAC and PostgreSQL's row-level security, and HIPAA compliance through AWS Security Hub. Chapter 3 provides definition of development methodology.

CHAPTER 3

RESULTS AND DISCUSSION

3.1 Equations

Use AMS packages when necessary. Here is an example of an equation array.

$$\begin{array}{rcl} A & = & B, \\ & = & D, \end{array} \quad (3.1)$$

$$E = F \quad (3.2)$$

3.2 Graphic and Images

A good thesis needs good diagrams/graphs/illustrations. Spend some time doing in properly. A good picture tells a thousand words.

3.3 Floats

- A float doesn't have a fixed location.
 - It can “float” forward or backward to wherever it fits best to get a high quality layout.
 - Caption as part of a float.
- Float Placement
 - h : try to place the float at the position where it is inserted
 - t : try to place the float at the top of the current page
 - b : try to place the float at the bottom of the current page
 - p : try to place the float at an own page



Figure 3.1 Example of a figure. This is a long, very long, long long, long caption. You can give a shorter caption for the “list of figures” using the square bracket symbol.

Table 3.1 The role of statistical quality engineering tools and methodologies

Temperature	Resonant Frequency	Q factor
13 mK $\pm$ 1 mK	16.93	811
40 mK $\pm$ 1 mK	16.93	817
100 mK $\pm$ 1 mK	16.93	815
300 mK $\pm$ 1 mK	16.93	806
500 mK $\pm$ 1 mK	16.93	811
800 mK $\pm$ 5 mK	16.93	814
1000 mK $\pm$ 5 mK	16.93	806

Table 3.2 The role of statistical quality engineering tools and methodologies

Temperature	Resonant Frequency	Q factor
13 mK $\pm$ 1 mK	16.93	811
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300 mK $\pm$ 1 mK	16.93	806
500 mK $\pm$ 1 mK	16.93	811
800 mK $\pm$ 5 mK	16.93	814
1000 mK $\pm$ 5 mK	16.93	806

Table 3.3 Feasible triples for highly variable Grid, MLMMH. Can include some context information (e.g. development software, test environment, procedure, limitations, assumptions, range of validity)

Time (s)	Triple chosen	Other feasible triples
0	(1, 11, 13725)	(1, 12, 10980), (1, 13, 8235), (2, 2, 0), (3, 1, 0)
2745	(1, 12, 10980)	(1, 13, 8235), (2, 2, 0), (2, 3, 0), (3, 1, 0)
5490	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
8235	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
10980	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
13725	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
16470	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
19215	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
21960	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
24705	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
27450	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
30195	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
32940	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
35685	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
38430	(1, 13, 10980)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
41175	(1, 12, 13725)	(1, 13, 10980), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
43920	(1, 13, 10980)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
46665	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
49410	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
52155	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
54900	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
57645	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
60390	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
63135	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
65880	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
68625	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
71370	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
74115	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)

Continued on next page

Table 3.3 – continued from previous page

Time (s)	Triple chosen	Other feasible triples
76860	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
79605	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
82350	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
85095	(1, 12, 13725)	(1, 13, 10980), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
87840	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
90585	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
93330	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
96075	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
98820	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
101565	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
104310	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
107055	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
109800	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
112545	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
115290	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
118035	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
120780	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
123525	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)

CHAPTER 4

CONCLUSION

That's all folks. Please read further on how to use the template. It is not a silver bullet.

4.1 Research Outcomes

4.2 Contributions to Knowledge

4.3 Future Works

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Appendix A Ransomware Simulation Time-series Data

Journal Articles

- (a) Paper 1
- (b) Paper 2

Conference proceedings

- (a) Paper 3
- (b) Paper 4